

FATE Monitoring: DOC Data - May want to rerun these samples

Winter 2025

2026-04-25

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##Setup - Change things here & write any notes

```
#identify section  
cat("Setup Information")
```

Setup Information

```
##### Run information - PLEASE CHANGE  
Date_Run = "03/19/26" #Date that instrument was run  
Run_by = "Stephanie J. Wilson" #Instrument user  
Script_run_by = "Stephanie J. Wilson" #Code user  
run_notes = "Check standards and duplicates were off on this run." #any notes from the run  
samples <- c("FATE") #whatever identifies your samples within the same names  
samples_pattern <- paste(samples, collapse = "|")  
  #samples_pattern <- "GCW" #use this instead of the line above if you have only one site code  
chks_name = "Chk_Std_50C_2N" #what did you name your check standards?  
  
##### File Names - PLEASE CHANGE  
#file path and name for raw summary data file  
raw_file_name = "Raw Data/FATE_DOC_202603.txt"  
  
#file path and name for raw all peaks file  
raw_allpeaks_name = "Raw Data/FATE_DOC_202603_allpeaks.txt"
```

```

#file path and name of processed data file
processed_file_name = "Processed Data/FATE_Processed_DOC_202601.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
#Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2025.csv"

#qaqc log file path for this year
Log_path = "Raw Data/FATE_TOCTN_QAQClog_2025.csv"

```

##Set Up Code

0.1 Import Data Functions

0.2 Import Sample Data

Import Sample Data

New names:

* ' -> '...14'

A tibble: 6 x 4

##	sample_name	npoc_raw	tdn_raw	run_datetime
##	<chr>	<dbl>	<dbl>	<chr>
## 1	FATE_Dock_20251002	5.97	0.237	3/19/2026 12:27:31 AM
## 2	FATE_Flume_20251002	7.38	0.235	3/19/2026 1:02:04 AM
## 3	FATE_Dock_20251103	6.01	0.161	3/19/2026 1:37:03 AM
## 4	FATE_Flume_20251103	8.51	0.276	3/19/2026 2:11:52 AM
## 5	FATE_Flume_20251103_dup	7.38	0.334	3/19/2026 2:44:35 AM
## 6	FATE_Dock_20251110	5.54	0.146	3/19/2026 3:19:28 AM

0.3 Assessing Standard Curves

Assess the Standard Curves

New names:

* ' -> '...18'

Warning: One or more parsing issues, call 'problems()' on your data frame for details,

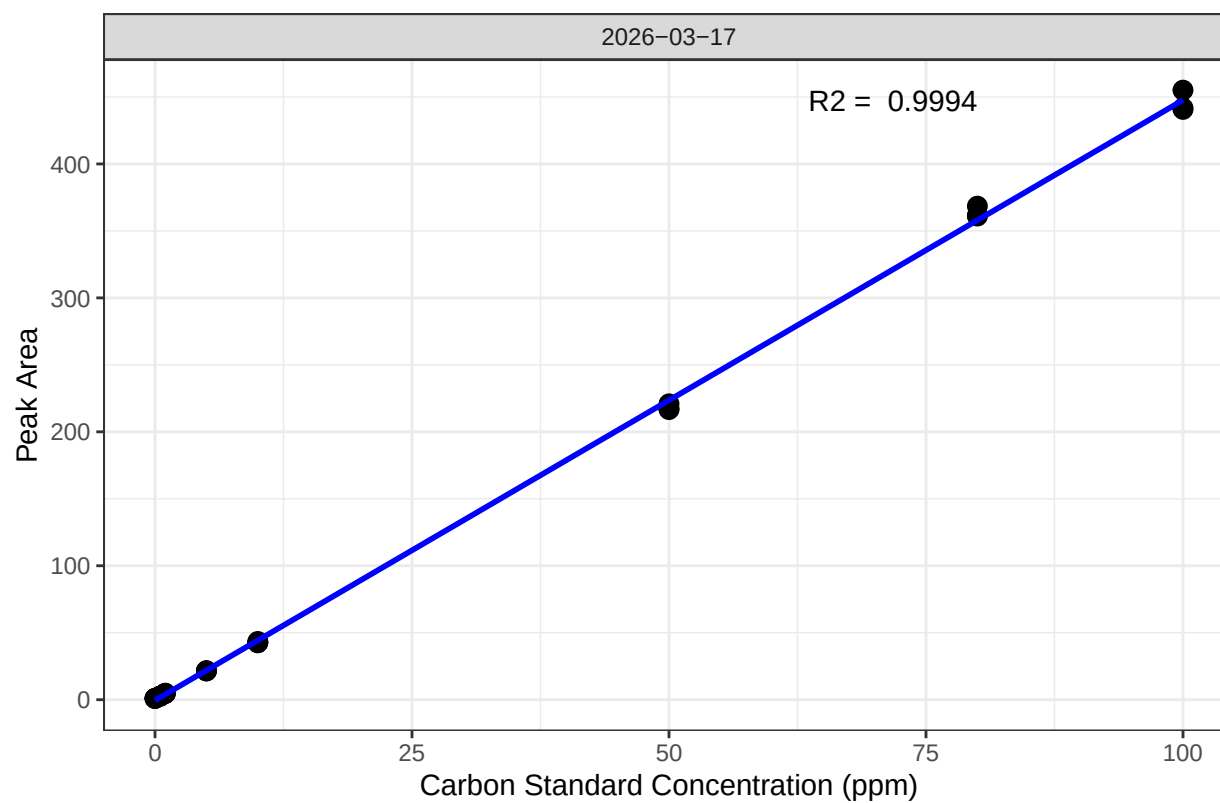
e.g.:

dat <- vroom(...)

problems(dat)

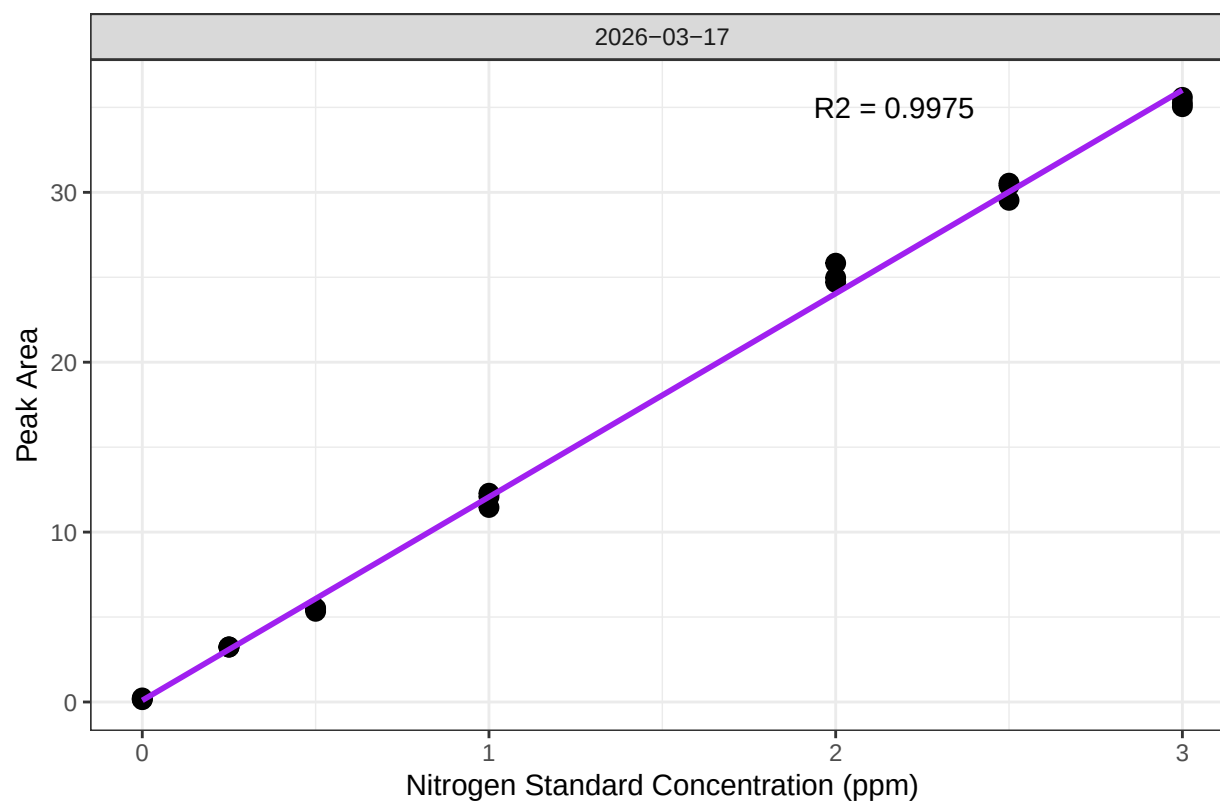
'geom_smooth()' using formula = 'y ~ x'

NPOC Std Curve by Date



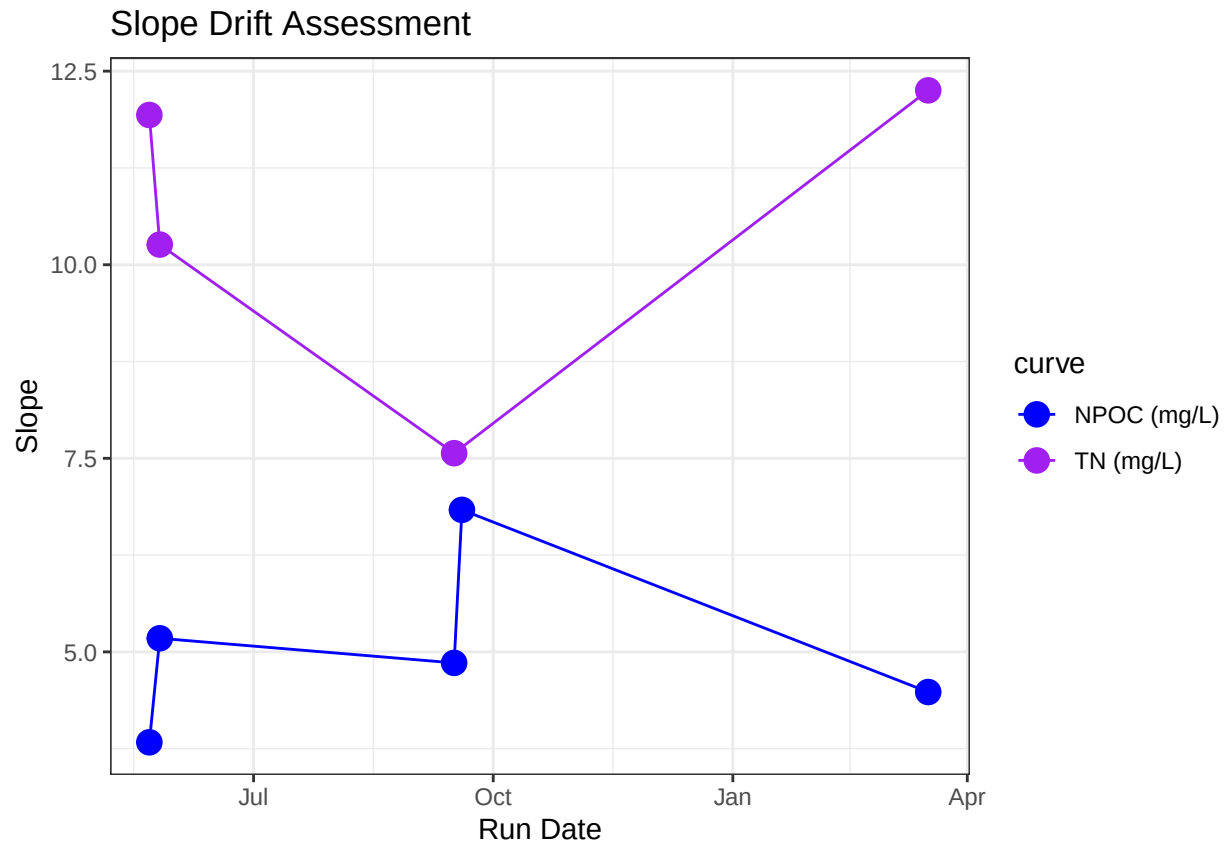
```
## 'geom_smooth()' using formula = 'y ~ x'
```

TN Std Curve by Date



```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_line()').
```



```
## [1] "NPOC Curve r2 GOOD"
```

```
## [1] "TN Curve r2 GOOD"
```

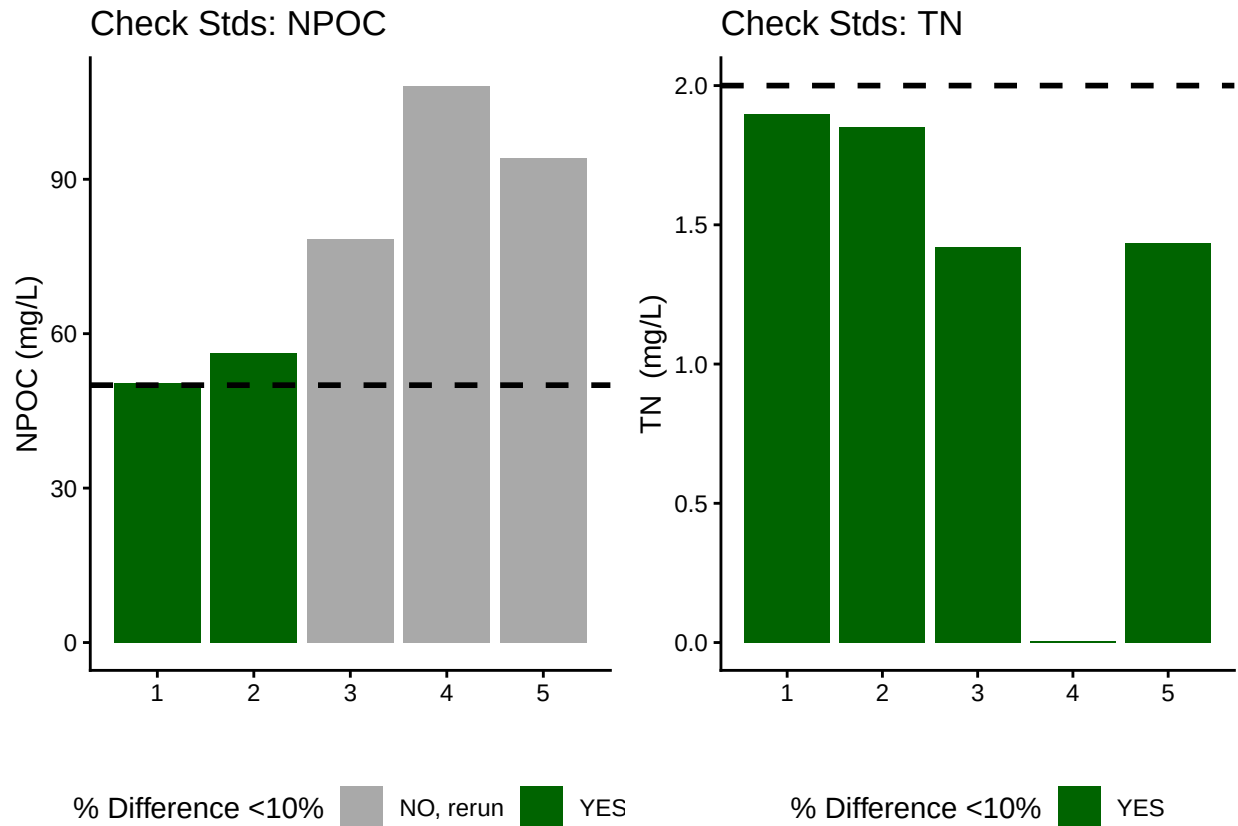
0.4 Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] "Carbon CHECK STANDARD RSD TOO HIGH - REASSESS"
```

```
## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```



```
## [1] ">60% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">60% of Nitrogen Check Standards are within range of expected concentration"
```

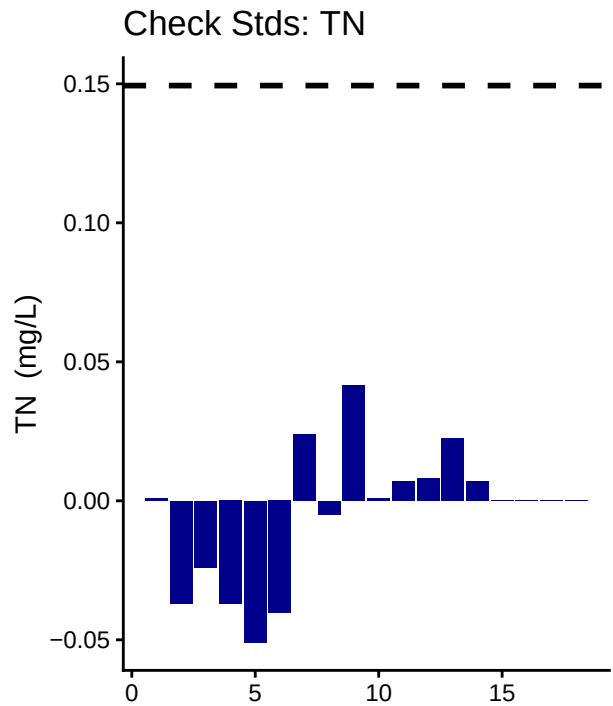
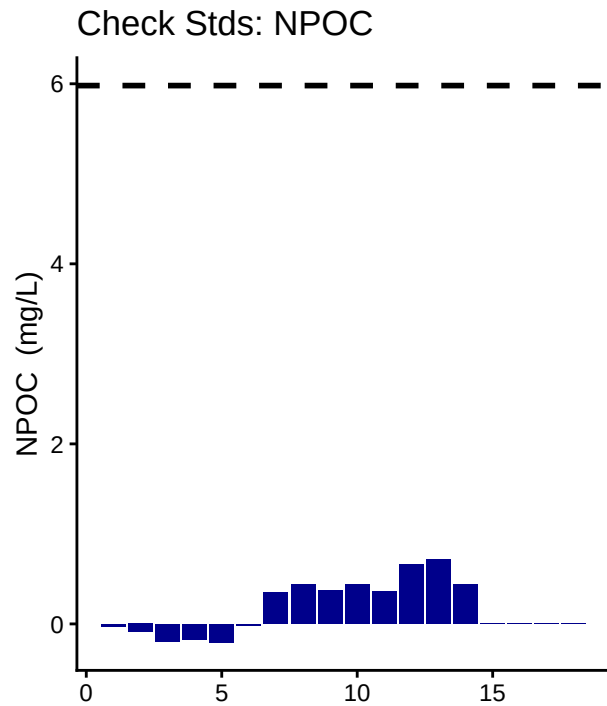
0.5 Assess Blanks

```
## Assess Blanks
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">60% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples ■ YE

Blank Conc <25% Quartile Samples ■ Y

```
## carbon blanks:
```

```
## [1] 0.1713417
```

```
## nitrogen blanks:
```

```
## [1] -0.004562222
```

0.6 Assess Duplicates

```
## Assess Duplicates
```

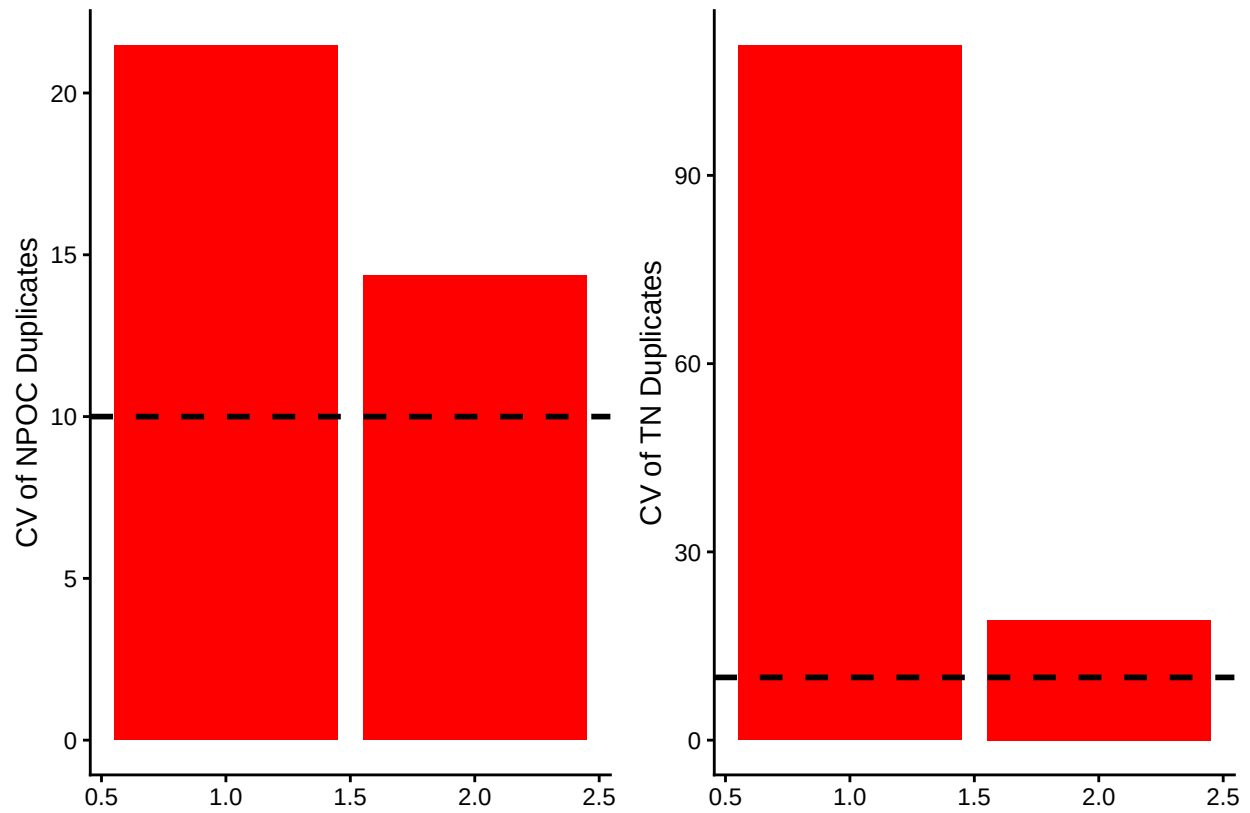
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
```

```
## i Please use 'linewidth' instead.
```

```
## This warning is displayed once per session.
```

```
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
```

```
## generated.
```

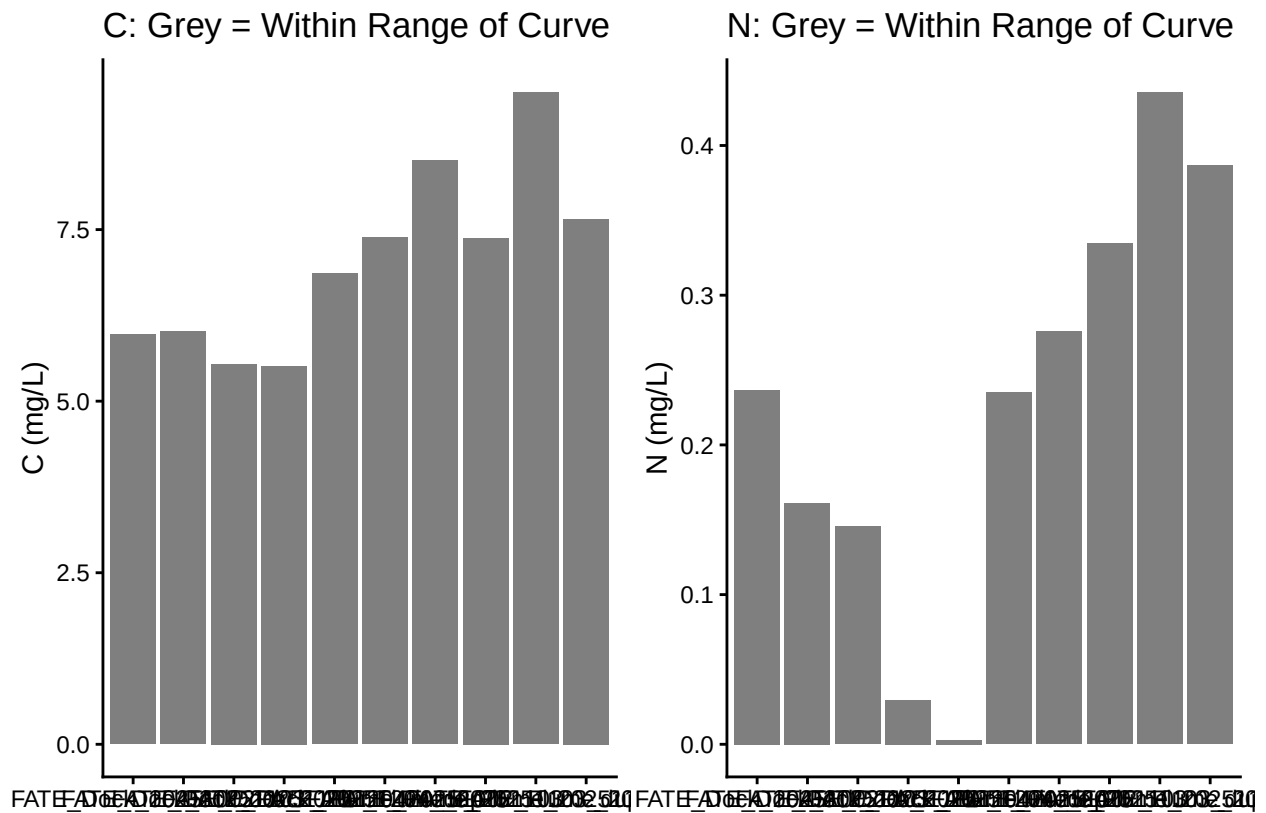


```
## [1] "<60% of Carbon Duplicates have a CB <10% - REASSESS"
```

```
## [1] "<60% of Nitrogen Duplicates have a CB <10% - REASSESS"
```

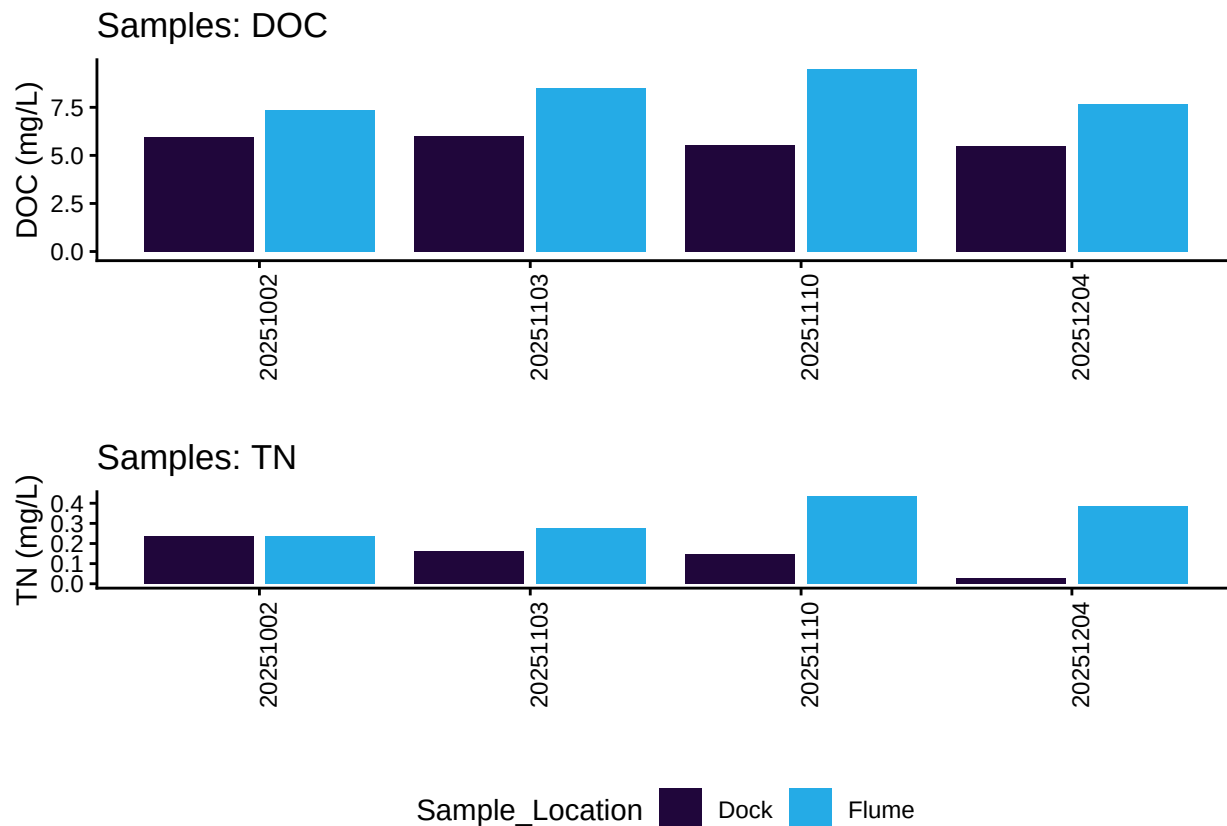
0.7 Sample Flagging - Are samples Within the range of the curve?

```
## Sample Flagging
```

0.8 Visualize Data by Plot

```
## Visualize Data
```



0.9 Convert data from mg/L to uMoles/L

0.10 Export Processed Data

Export Processed Data

```
## # A tibble: 6 x 11
##   Project Sample_Location Date      npoc_mgL npoc_uM npoc_flag      tdn_mgL tdn_uM
##   <chr>    <chr>          <chr>      <dbl>    <dbl> <chr>      <dbl>  <dbl>
## 1 FATE    Dock            20251002    5.97    497. NPOC checks ~ 0.237  16.9
## 2 FATE    Flume           20251002    7.38    615. NPOC checks ~ 0.235  16.8
## 3 FATE    Dock            20251103    6.01    501. NPOC checks ~ 0.161  11.5
## 4 FATE    Flume           20251103    8.51    708. NPOC checks ~ 0.276  19.7
## 5 FATE    Dock            20251110    5.54    462. NPOC checks ~ 0.146  10.4
## 6 FATE    Flume           20251110    9.5     791. NPOC checks ~ 0.436  31.1
## # i 3 more variables: tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>
```

#end